# Global food security and fertilizer

"Surging fertilizer costs and wasteful spreading methods are forcing farmers to use less nutrients, lowering crop yields and threatening global food security." *Generated by LeapSpace*

## Bottom line

**Yes, the core claim is supported: higher fertilizer costs and inefficient spreading push farmers to cut nutrient use, and that can lower yields and threaten food security—especially where farmers cannot absorb the price shock or where nutrient management is already inefficient.** At the same time, the evidence shows this is **not inevitable**: precision nutrient management, site-specific fertilization, and enhanced-efficiency fertilizers can maintain or raise yields while reducing inputs and losses. [1] [2] [3] [4] [5] [6] [7]

## What the evidence says

### 1) Fertilizer price spikes can reduce use, but effects are uneven

Global fertilizer price shocks raised fears of lower application, lower production, and worse food insecurity; the review on recent market shocks found that global fertilizer demand fell only modestly overall, but the burden was heavier for many farmers in Africa. [1] In Africa, the crisis led to two- to threefold local price increases and farm-level responses such as reducing fertilizer quantities and cropped area, which were expected to reduce productivity. [8] A separate Egypt case study likewise reports a 20.9% decline in nitrogen fertilizer use per hectare from 2018 to 2024 and links high input costs and volatility to the need for subsidies and context-specific policy. [9]

### 2) Under-application and inefficient application can reduce yields

Multiple studies link insufficient or imbalanced fertilization with lower yields and yield gaps. Soil nutrient depletion is described as reducing crop yield and threatening food security globally, while unbalanced fertilization and nutrient deficits are identified as major causes. [10] In wheat systems, reduced nutrient input led to inadequate soil nutrient supply, limited growth, and lower yield and nutrient-use efficiency. [11] More broadly, reviews note that overreliance on nitrogen fertilizer has also produced declining nutrient-use efficiency, crop productivity decline, soil acidification, and nitrate contamination. [12] [13]

### 3) Better nutrient management can offset the problem

The evidence consistently shows that **smarter fertilizer use is the key lever**. Precision nutrient management, site-specific nutrient management, and optimized timing improve nutrient-use efficiency and often maintain or increase yields while reducing losses. [6] [7] [14] [15] [16] Field studies report yield gains or maintained yields with lower inputs under optimized fertilization, including vegetable systems, rice systems, and maize systems. [7] [17] [18] [19] [20] Reviews of enhanced-efficiency fertilizers and next-generation fertilizer technologies also conclude they can raise NUE and reduce environmental burden. [5] [21] [22]

## Synthesis

The strongest reading of the evidence is that **rising fertilizer costs and wasteful application methods do not just squeeze farm budgets; they also create a yield and food-security risk when farmers respond by underapplying nutrients or when nutrients are used inefficiently.** But the same literature also shows that **the path to protecting yields is not more blanket fertilizer use—it is better targeting, better timing, and better formulations.** [1] [3] [4] [6] [7]

Confidence level: **High** — the retrieved studies and reviews consistently support the link between fertilizer cost/inefficiency, reduced nutrient use, yield risk, and food-security pressure, while also consistently showing that precision and optimized nutrient management can mitigate those harms.

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